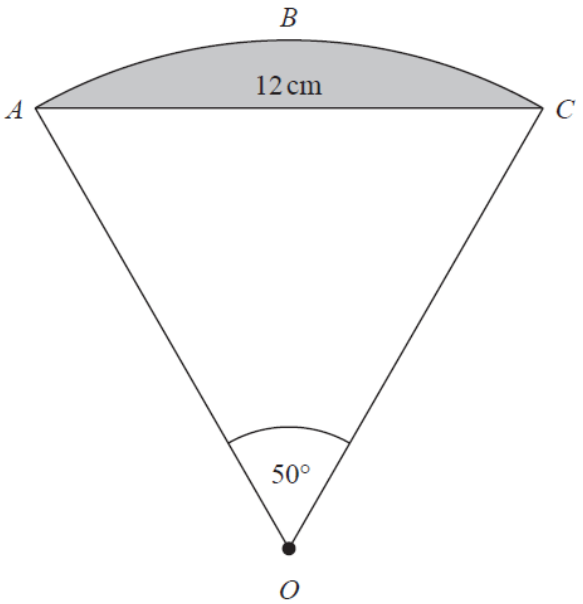
20 *OABC* is a sector of a circle with centre *O*.



Angle *AOC* = 50°
AC = 12 cm

Work out the area of the shaded segment of the circle.
Give your answer correct to 3 significant figures.

Bisect the sector: half-angle 25°, half-chord 6 cm.

1 mark
Use half of AC.

$r = 6 \div \sin 25° = 14.19720950...$ cm

1 mark
Find radius.

Sector area = $\frac{50}{360} \times \pi r^2 = 87.94747...$

1 mark
Sector formula.

1 mark
Triangle formula.

Triangle = $\frac{1}{2}r^2 \sin 50° = 77.20224914...$

Segment = sector – triangle = 10.74522...

= 10.7 cm^2 (3 s.f.)

1 mark
10.7 cm^2.

..... cm^2

(Total for Question 20 is 5 marks)